Exponentially Many Barnette Graphs with the Global H^{+-} Property

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Abstract

A Barnette graph is simple, cubic, 3-connected, bipartite and planar. It has the global H^{+-} property if, for every ordered pair of distinct edges (e, f), some Hamilton cycle contains e and avoids f. Barnette's conjecture has been proved (OpenAI, 2026; exposition by Z. Li, arXiv:2610.30490), and with it P_2-, P_4-Hamiltonicity, H^- and the H^{+-} condition for edge pairs at distance at most 1 hold for every Barnette graph; global H^{+-} for all Barnette graphs remains open. All Barnette graphs on at most 66 vertices are known computationally to have it, and infinite families follow from tight-cut composition. We give a controlled expansion mechanism with an explicit growth rate. Starting from the cube, we iterate legal C_4-expansions at two recurrent sites. An exact six-traversal criterion describes the Hamilton cycles of one expansion in terms of parent structures. A finite signed-support invariant (22 states, or 40 when reflected orientations are allowed), initialised by explicit cube witnesses, then shows that every construction word yields a Barnette graph with global H^{+-}. A rooted grammar argument shows that depth k produces exactly $2^{\max(k-2,0)}$ boundary-pointwise isomorphism types. A cubic embedding bound turns this into $B_{H^{+-}}(8+4k) \geq \lceil 2^{\max(k-2,0)}/((8+4k)3^7)\rceil$, so $B_{H^{+-}}(n) \geq 2^{n/4-O(\log n)}$ along $n \equiv 0 \pmod 4$. Exhaustive computer checks agree with these statements for all standard words with $n \leq 32$ and all reflected words with $n \leq 28$. Limitations: the result concerns two specific expansion sites, not arbitrary C_4-expansions. The explicit bound first exceeds 1 at $n = 88$. A 2005 edge-replacement scheme of Kao and Chen could not be consulted in full, so we make no absolute priority claim.

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Problem and contribution

Barnette's conjecture asserts that every Barnette graph (simple, cubic, 3-connected, bipartite and planar) is Hamiltonian. It has now been proved: OpenAI [1] (preprint dated 24 September 2026, released 6 October 2026) proves Hamiltonicity (Theorem 1.1), shows that every edge is avoided by some Hamilton cycle (the H^- property, Corollary 7.1), and, through Theorem 4.16 of [2], that every path with three edges lies in a Hamilton cycle (P_4-Hamiltonicity, Corollary 7.2); Li [3] gives an independent exposition of the proof and shows that $G - e$ is Hamiltonian for every Barnette graph G and edge e (Theorem 3). Consequently every Barnette graph is P_2-Hamiltonian and has H^- and P_4. It also satisfies the H^{+-} condition for every ordered pair (e, f) of distinct edges at distance at most 1, where the distance between two edges is defined as the minimum graph distance between their endpoints (so incident edges have distance 0): if $e = ab$ and $f = bc$ share b, let d be the third neighbour of b (so $d \notin \{a, c\}$); any Hamilton cycle through the two-edge path a-b-d (which exists by P_4-Hamiltonicity) contains e and avoids f, and if $f = cd$ is at distance 1 from $e = ab$, joined by the edge bc, any Hamilton cycle through a three-edge path a-b-c-z with z the third neighbour of c outside $\{b, d\}$ contains e and avoids f. Several stronger Hamiltonicity properties are tied to it. Kelmans proved that the conjecture is equivalent to every Barnette graph having the facial H^{+-} property: for any two edges on a common face there is a Hamilton cycle through one and avoiding the other (as stated in [4]). The *global* H^{+-} property used here, following Gorsky, Steiner and Wiederrecht [2], asks this for every ordered pair of distinct edges, with no facial restriction. Computer search shows that all Barnette graphs on at most 66 vertices have global H^{+-} (Brinkmann, Goedgebeur and McKay [5], as reported in [2], Theorem 5.7). Infinite families with global H^{+-} already follow from the tight-cut composition theorem of [2] (Lemma 4.5). Whether every Barnette graph has global H^{+-} remains open; neither [1] nor [3] claims it. What this paper adds is a controlled expansion mechanism, giving explicit recursive families with global H^{+-}: explicit words in two C_4-expansion sites, an exact local transfer law along them, and an explicit exponential growth rate for the number of graphs obtained.

Unconditional results.

1. (Main theorem, Theorem 1.1.) Let $B_{H^{+-}}(n)$ be the number of isomorphism classes of n-vertex Barnette graphs with global H^{+-}. For every $k \geq 0$,
$$B_{H^{+-}}(8+4k) \geq \left\lceil \frac{2^{\max(k-2,0)}}{(8+4k)\,3^7} \right\rceil,$$
so $B_{H^{+-}}(n) \geq 2^{n/4-O(\log n)}$ along $n \equiv 0 \pmod 4$, $n \geq 8$. The graphs are explicit: they are produced from the cube by words in two legal C_4-expansion sites, called CEN and UV.
2. (Transfer criterion, Theorem 8.2.) A finite signed-support invariant (22 states for the two standard sites, 40 states when reflected orientations are allowed) together with a joint-separation condition at a root site implies global H^{+-} for every descendant along every word. The criterion applies to any root satisfying the stated hypotheses; the cube is verified by explicit witnesses.
3. (Local tools.) An exact six-traversal criterion for Hamiltonicity of one C_4-expansion (Theorem 3.1), the seven inclusion-minimal traversal covers of the eight new edges (Theorem 4.1), and a rooted classification of the construction patches: at depth k there are exactly $2^{\max(k-2,0)}$ boundary-pointwise isomorphism types (Theorem 9.2).

The main contribution is item 1; items 2 and 3 are the steps of its proof. Section 5, Theorem 7.1 and Corollary 7.2 give the weaker edge-Hamiltonicity (P_2-Hamiltonicity) version of the same mechanism. Global H^{+-} implies it, and by [1] it holds for all Barnette graphs, so they are kept only because the later sections use their notation and tables.

Not claimed. While Lemma 2.2 and [1], [3] imply that the output of any legal C_4-expansion of a Barnette graph is Hamiltonian and P_4-Hamiltonian (hence P_2-Hamiltonian) at the graph level, we do not claim that our local witness-transfer mechanism applies to an arbitrary legal C_4-expansion site, or that an arbitrary legal C_4-expansion preserves global H^{+-}; nor do we claim that the 2^k construction histories at depth k give non-isomorphic graphs (they do not), that the cube core can be recognized in an unlabelled descendant, or that global H^{+-} holds for all Barnette graphs. The explicit bound is weak for small orders: it first exceeds 1 at $n = 88$.

Related work and distinction

Generation and C_4-expansion. Holton, Manvel and McKay [6] generate Barnette graphs by cube- and C_4-expansions and use facial H^{+-}-type properties in their computer verification. Gorsky, Steiner and Wiederrecht [2] show that H^{+-} is preserved under tight-cut decomposition and reduces to braces. They also note that H^{+-} itself is not preserved by the generation procedure. Their Lemma 4.5 shows that a cubic 3-connected bipartite graph with a nontrivial tight cut has H^{+-} if and only if both tight cut contractions do, so gluing graphs with H^{+-} along tight cuts already gives infinite families with global H^{+-}. Our construction does not use tight cuts. It stays inside braces: by [2], Lemma 5.2, a C_4-expansion of a 2-extendable bipartite graph is 2-extendable, and the cube is 2-extendable, so every graph in our family is a brace, the setting to which [2] reduces the H^{+-} question. Although Lemma 2.2 and [1], [3] imply that every legal C_4-expansion of a Barnette graph is Hamiltonian and P_4-Hamiltonian at the graph level, a specific parent witness need not lift across an arbitrary site and global H^{+-} need not be preserved; in a constructive setting without the general theorem, Bej [7] likewise distinguishes structured recursive square insertions in annular-decomposable Barnette graphs from arbitrary repetition. Our expansion operation is the standard one. What is new is the exact parent-side criterion for one specified site, the finite signed-support invariant that carries the ordered contain-one/avoid-one property along two specific recurrent sites, and the resulting explicit count. The contribution is the controlled mechanism and its growth rate, not the existence of an infinite family with global H^{+-}.

Strengthened Hamiltonicity in Barnette graphs. Kelmans' equivalence and Florek's results [4] concern the facial H^{+-} property (edges on a common face). Florek proves it, in a dual formulation, for even triangulations whose high-degree colour classes induce forests. Prescribed-edge Hamiltonicity is discussed as a strengthening by Hertel [8]. Hamiltonicity is known for Barnette graphs whose faces have size 4 or 6 (Goodey [9]), and for faces of size up to 8 (Schnieders [10]). Related classes include planar cubic graphs with faces of size at most 6 (Kardoš [11]) and Barnette graphs with few big faces adjacent to any face (Florek [12]). These results, and the general proof [1], [3], give Hamiltonicity, H^{-}, P_4 or facial properties. Our property is the global ordered-pair property, on a much narrower explicit class. The finite verification of [5] covers all Barnette graphs up to 66 vertices but gives no infinite family.

Infinite highly Hamiltonian bipartite planar cubic families. Spider-web networks [13], [14] are infinite families with strong fault-tolerant and laceability properties. They have one graph per order, which does not give an exponential count. Kao and Chen [15] describe an edge-replacement scheme that generates bipartite graphs with fault-tolerant Hamiltonian properties. Its indexed abstract states the conclusions precisely: the generated graphs stay Hamiltonian

after deleting one edge or one pair of vertices of opposite colours, and are Hamiltonian laceable. We could not legally obtain its full text, and the abstract does not describe the replacement gadget. Whether a specialization of that scheme contains the present mechanism is therefore undecided. For this reason we make no absolute priority claim. The novelty statement is limited to: in searches of OpenAlex, Crossref and arXiv (25 October 2026), we found no explicit exponential lower bound for the number of Barnette graphs with global H^{+-}.

rStation. This is a revision of the accepted version of this paper on the `math` board; no other paper on that board overlaps with it.

1. Introduction

A *Barnette graph* is a finite simple cubic 3-connected bipartite planar graph. Barnette's conjecture, that every such graph is Hamiltonian, has now been proved [1], [3]. We study a stronger property on an explicitly constructed subclass. A graph has the *global H^{+-}-property* when, for every ordered pair of distinct edges (e, f), it has a Hamilton cycle containing e and avoiding f. This is stronger than both P_2-Hamiltonicity, where every edge belongs to a Hamilton cycle, and H^{-}, where every edge is avoided by a Hamilton cycle. The terminology follows Gorsky, Steiner and Wiederrecht [2]. In particular, P_2 is the two-vertex path, so P_2-Hamiltonicity is prescribed-edge Hamiltonicity.

Holton, Manvel and McKay introduced cube- and C_4-expansions in their generation theory for Barnette graphs [6], [2]. The operation itself is therefore standard. Within the Barnette class, Lemma 2.2 and the general theorems of [1], [3] imply that the output of every legal C_4-expansion is again Hamiltonian and P_4-Hamiltonian at the graph level, but a given parent witness need not lift through an arbitrary expansion site, nor does an arbitrary legal expansion preserve global H^{+-} [2]. Bej's analysis of annular-decomposable Barnette graphs similarly distinguishes structured recursive applications from arbitrary repetition of the corresponding square insertion at the level of explicit cycle constructions [7]. Our starting point is consequently local and witness-explicit: for one specified legal expansion site, which Hamiltonian structures of the parent lift to Hamilton cycles of the expansion, how do they cover the new edges, and which support data survive a recursively chosen sequence of sites?

The local answer has three layers. First, a Hamilton cycle in the expanded graph is equivalent to one of six parent traversals: three Hamilton-cycle states, two cross Hamilton paths, or a spanning two-path linkage with the correct endpoint pairing. Second, the eight gadget edges are all Hamiltonian exactly when the available traversal types contain one of seven inclusion-minimal pairs. Third, retaining individual old edges requires memberwise support rather than the mere availability of role names. This gives a finite edge-cover transfer. Global H^{+-} requires more: for each retained edge we track roles of concrete witnesses that use it and roles of witnesses that avoid it, while retaining a same-witness separation condition for ordered pairs of retained edges.

Two recurrent sites are decisive. After one expansion, CEN selects the opposite central edges bc, da, while UV selects the two spokes ua, bv on the adjacent external face. Splitting the three four-terminal pairings gives a complete planar boundary interface. On the states reachable from the cube, this reduces to six roles. The two transfer relations are nonconjugate, and the signed support of every retained edge stays in an explicit 22-state closed set for the standard orientation. Including the reflected orientation gives a 40-state forward closure. The local incidence tables also preserve a common separating witness for every ordered pair of retained edges. Eight cube witnesses initialize the invariant. Hence every finite word in $\{\mathrm{CEN}, \mathrm{UV}\}$ produces a Barnette graph with global H^{+-}. Allowing ordered-terminal reflections uses the 40-state forward closure described below; the quantitative count uses only the standard subgrammar.

Construction histories need not be nonisomorphic. We avoid inferring unlabelled growth directly from the binary history tree. Instead, we first classify the graphs with the old labelled core fixed pointwise. The ordered-boundary four-pole grammar has a two-step delay: every letter except the last effective expansion choice is recoverable by a local adjacency test, whereas the two final eight-vertex caps are boundary-pointwise isomorphic. This gives exactly $2^{\max(k-2,0)}$ rooted types at depth k. We then forget the root by a coarse but sufficient orbit count.

For $n \geq 0$ let $B_{H^{+-}}(n)$ denote the number of isomorphism classes of n-vertex Barnette graphs with global H^{+-}. Our main quantitative statement is the following.

Theorem 1.1. For every integer $k \geq 0$,

$$B_{H^{+-}}(8 + 4k) \geq \left\lceil \frac{2^{\max(k-2,0)}}{(8 + 4k)\, 3^7} \right\rceil,$$

where the denominator is the cubic marking bound. In particular, along the orders $n = 8 + 4k$,

$$B_{H^{+-}}(n) \geq 2^{n/4-\log_2 n-4-7\log_2 3} = 2^{n/4-O(\log n)}.$$

The theorem counts graphs, not Hamilton cycles inside one graph. It does not assert that all 2^k histories at depth k are nonisomorphic, that the old cube core is uniquely recognizable in an unlabelled descendant, or that an arbitrary legal C_4-expansion preserves global H^{+-}.

2. The legal C_4-expansion

All graphs are finite and simple. A Hamilton path is spanning. A spanning two-path linkage is a pair of vertex-disjoint positive-length paths whose vertex sets partition the graph.

Definition 2.1. Let G be a graph and let

$$p = uv, \qquad q = xy$$

be vertex-disjoint edges. The C_4-expansion of G at the ordered site (u, v, x, y) is the graph H obtained by deleting p, q, adding vertices a, b, c, d, and adding the eight edges

$$ua,\ ab,\ bv,\ xc,\ cd,\ dy,\ ad,\ bc. \tag{2.1}$$

For a plane bipartite graph, the site is *legal* when p, q lie on a common face in the cyclic orientation

$$u, v, \dots, x, y, \dots, u \tag{2.2}$$

and u, x are in the same bipartition class.

Lemma 2.2. A legal C_4-expansion of a Barnette graph is again a Barnette graph.

Proof. The order in (2.2) permits the square and its four spokes to be drawn inside the common face without crossings. Cubicity and simplicity are immediate from (2.1). If u, x receive one bipartition colour, then a, c receive the other colour and b, d receive the original colour; every edge in (2.1) is bichromatic.

It remains to check 3-vertex-connectivity. Let $S \subseteq V(H)$ with $|S| \leq 2$, and regard a, b, c, d as the internal square. Every component of $H - S$ contains an old vertex: after at most two deletions, a component of the internal square cannot lose both of its square neighbours and its distinct old attachment.

If S contains no internal vertex, connectivity of $G - (S \cap V(G))$ and the connected square reroute every surviving use of p or q. If S contains two internal vertices, then it contains no old vertex; $G - \{p, q\}$ is connected because a cubic 3-vertex-connected graph is 3-edge-connected. Suppose finally that S contains one internal vertex and one old vertex z. The graph $G - z$ is 2-connected. If z is an endpoint of p or q, that edge is already absent from $G - z$; deleting the other site edge cannot disconnect the 2-connected graph. Thus, whenever $(G - z) - \{p, q\}$ is disconnected, both site edges survive and form a two-edge cut, so each of uv and xy has endpoints on opposite sides. Deleting one internal square vertex leaves a path joining three of the four attachments; any three of u, v, x, y meet both sides of that two-edge cut. The surviving square path therefore reconnects the sides. Thus $H - S$ is connected in every case. □

3. The exact six-traversal interface

Put $R = G - \{p, q\}$, deleting the two edges but retaining all vertices. We use the following parent structures:

$$\begin{aligned}
C_{11} &: \text{ a Hamilton cycle containing both } p \text{ and } q,\\
C_{10} &: \text{ a Hamilton cycle containing } p \text{ and avoiding } q,\\
C_{01} &: \text{ a Hamilton cycle containing } q \text{ and avoiding } p,\\
X_{vx} &: \text{ a Hamilton } v\text{-}x \text{ path in } R,\\
X_{uy} &: \text{ a Hamilton } u\text{-}y \text{ path in } R,\\
L &: \text{ a spanning two-path linkage in } R \text{ with endpoint set } \{u, v, x, y\}.
\end{aligned}$$

For L, the endpoint pairing will be specified below.

Write the gadget-edge incidence in the order

$$(ua, ab, bv, xc, cd, dy, ad, bc). \tag{3.1}$$

The six nonisolated local patterns are shown in Table 1. The symbol D_{ij} denotes the child traversal arising from C_{ij}.

type	parent witness	incidence vector	missing gadget edges
D_{11}	C_{11}	11111100	ad, bc
D_{10}	C_{10}	10101011	ab, xc, dy
D_{01}	C_{01}	01010111	ua, bv, cd
X_{vx}	Hamilton v-x path	01111010	ua, dy, bc
X_{uy}	Hamilton u-y path	11001101	bv, xc, ad
L	spanning two-path linkage	10110111	ab, cd

Table 1. The six realizable traversals of one C_4-expansion.

Theorem 3.1 (six-traversal criterion). The expanded graph H is Hamiltonian if and only if at least one of the following holds: (i) C_{11} exists;

(ii) C_{10} exists;

(iii) C_{01} exists;

(iv) X_{vx} exists;

(v) X_{uy} exists;

(vi) R has a spanning two-path linkage whose endpoint pairing is not $(uy)(vx)$.

Proof. Let K be a Hamilton cycle of H and intersect K with the eight edges in (2.1). Imposing degree two at each of a, b, c, d gives seven subsets. One is the isolated square $abcda$; it cannot be a component of a Hamilton cycle containing the old vertices. The remaining six subsets are exactly the incidence rows in Table 1. Removing their gadget edges leaves, respectively, the three Hamilton-cycle states, the two cross Hamilton paths, and a spanning two-path linkage on the old vertices. This proves necessity apart from the connectivity condition for L.

In the four-spoke case, both the gadget part and the old linkage are unions of two paths. Contract each path to an edge. The gadget induces the perfect matching $(uy)(vx)$ on $\{u, v, x, y\}$. The union is one cycle exactly when the old and gadget matchings are distinct; equal matchings give two cycles. This proves the exclusion in (vi).

Conversely, adjoin to any witness in (i)–(vi) the corresponding row of Table 1. Every vertex then has degree two. In the linkage case the preceding matching argument gives one component; in the other five cases connectivity is immediate from the spanning parent cycle or path. The resulting spanning 2-regular connected graph is a Hamilton cycle of H. □

Remark 3.2. Theorem 3.1 uses neither planarity nor bipartiteness. Those hypotheses enter only when the expansion is required to remain in the Barnette class. The linkage pairing condition cannot be replaced by degree equations: the excluded pairing gives two disjoint cycles. The two other parent-linkage pairings both realize the single local incidence type L in Table 1; they need not be split for gadget-edge coverage.

4. Covering the eight gadget edges

A traversal type is *available* when the corresponding parent witness exists. Every Hamilton cycle of the child has exactly one of the six patterns in Table 1.

Theorem 4.1 (minimal gadget covers). Every edge of the installed gadget lies on a Hamilton cycle of H if and only if the available traversal types contain at least one of the following seven pairs:

$$\begin{gathered}\{D_{11}, D_{10}\}, \quad \{D_{11}, D_{01}\}, \quad \{D_{10}, D_{01}\}, \quad \{D_{11}, L\}, \\ \{X_{vx}, X_{uy}\}, \quad \{X_{vx}, L\}, \quad \{X_{uy}, L\}.\end{gathered} \tag{4.1}$$

These are precisely the inclusion-minimal covers.

Proof. For each row of Table 1, let M_W be its set of missing edges. Two rows cover the gadget exactly when their missing sets are disjoint. Comparing the six displayed missing sets gives precisely the seven pairs in (4.1). No single row covers all eight edges.

It remains only to exclude an inclusion-minimal cover of size at least three. The graph whose vertices are the six types and whose edges join pairs with intersecting missing sets is triangle-free: this follows directly from the six missing sets in Table 1. Hence every three traversal types contain two with disjoint missing sets, and therefore contain one of the seven covering pairs. A larger cover cannot be inclusion-minimal either. □

The theorem separates the new-edge problem from the retained-edge problem. Availability is Boolean for the eight gadget edges. To iterate an expansion while keeping every earlier edge Hamiltonian, one must follow which concrete witnesses contain each retained edge.

5. Memberwise support and a five-role robust domain

Set

$$A = C_{11}, \qquad B = C_{10}, \qquad C = C_{01}, \qquad D = X_{vx}, \qquad E = X_{uy}. \tag{5.1}$$

For a family of witnesses in a role W, write $E(W)$ for the union of their edge sets. For a retained edge g, its support is

$$S(g) = \{W \in \{A, B, C, D, E\} : g \in E(W)\}. \tag{5.2}$$

The transfer is memberwise: the local construction is applied independently to each witness in a family, and every parent edge other than the two site edges is retained.

For the central successor site

$$(p^+, q^+) = (bc, da), \qquad (u^+, v^+, x^+, y^+) = (b, c, d, a), \tag{5.3}$$

the canonical origins of child roles are

$$A^+ \leftarrow B, \qquad B^+ \leftarrow E, \qquad C^+ \leftarrow D, \qquad D^+ \leftarrow A, \qquad E^+ \leftarrow A. \tag{5.4}$$

Thus a Boolean support vector transforms as

$$T(a, b, c, d, e) = (b, e, d, a, a). \tag{5.5}$$

Reflecting the orientation interchanges the two one-edge cycle states and the cross paths, and gives

$$\overline{T}(a, b, c, d, e) = (c, e, d, a, a). \tag{5.6}$$

Proposition 5.1 (greatest five-role robust domain). Let

$$\mathcal{U} = \{S \neq \varnothing : A \in S \text{ or } \{B, C\} \subseteq S \text{ or } \{D, E\} \subseteq S\}. \tag{5.7}$$

Then $\mathcal{U}$ is the greatest collection of nonempty support states invariant under both maps (5.5)–(5.6). Its minimal states are

$$\{A\}, \qquad \{B, C\}, \qquad \{D, E\}, \tag{5.8}$$

and either orientation sends these states around the cycle

$$\{A\} \longmapsto \{D, E\} \longmapsto \{B, C\} \longmapsto \{A\}. \tag{5.9}$$

Proof. Equation (5.9) follows immediately from (5.5)–(5.6); since $\mathcal{U}$ is the upward closure of the three states in (5.8), it is invariant.

Conversely, suppose $S \notin \mathcal{U}$. Then $A \notin S$, at least one of B, C is absent, and at least one of D, E is absent. Choose T if $B \notin S$, and choose $\overline{T}$ if $C \notin S$. The image has neither A nor both B, C, and it has no D, E support because A was absent.

It is therefore empty or a singleton in $\{B, C\}$. One further adversarial choice sends that singleton to the empty state. No invariant nonempty domain can contain S, proving maximality. □

For completeness, the eight newly installed edges also return to $\mathcal{U}$. In the order (3.1), their standard five-role output supports are

$$(ABDE,\ BCD,\ ACDE,\ CDE,\ ABCE,\ BDE,\ AC,\ AB). \tag{5.10}$$

Every entry lies in $\mathcal{U}$, and reflection preserves (5.7). This gives a sufficient support theorem for an arbitrary tree of reflected central orientations. It is not a statement that every legal C_4-expansion preserves P_2-Hamiltonicity.

6. The nonconjugate CEN/UV transfer

We now keep one central successor and replace the reflected copy by a geometrically different site. After the expansion (2.1), set

$$\mathrm{CEN} = (bc, da), \qquad \mathrm{UV} = (ua, bv). \tag{6.1}$$

The first site uses opposite edges of the central square. The second uses the two new spokes on the face obtained by replacing uv with $u - a - b - v$. Both sites satisfy Definition 2.1: in the orientations (b, c, d, a) and (u, a, b, v), respectively, the initial vertices of the two selected edges have the same bipartition colour.

The role $A = C_{11}$ must now be split according to the pairing left after deleting p, q:

$$A_0 : (uy)(vx), \qquad A_1 : (ux)(vy). \tag{6.2}$$

Together with B, C, D, E from (5.1), these give six interface roles.

Proposition 6.1 (two-site transfer). The memberwise source-to-target relations for the two sites are

	A_0^+	A_1^+	B^+	C^+	D^+	E^+
CEN	$B \vee C$	A_1	E	D	$A_0 \vee A_1$	$A_0 \vee A_1$
UV	$A_0 \vee A_1 \vee B$	A_1	E	D	C	$A_0 \vee A_1$.

(6.3)

Every source column is nonzero for both relations. Consequently the greatest common domain in which one retained edge has nonempty support under every word in the two relations is

$$2^{\{A_0, A_1, B, C, D, E\}} \setminus \{\varnothing\}. \tag{6.4}$$

Proof. Delete the two parent site edges from one source witness and contract every external path segment that contains no terminal. For A_0, A_1, this leaves the pairings

$$A_0 = (uy)(vx), \qquad A_1 = (ux)(vy).$$

Representative contracted paths for the other four roles are

$$B : u - x - y - v, \qquad C : x - u - v - y, \qquad D : v - u - y - x, \qquad E : u - v - x - y.$$

Interchanging the two internal terminals of any one of these paths does not change the case split: those terminals already have degree two, so their incident gadget spokes are forced to be absent. Thus the displayed paths also cover the reverse inherited orders. A target witness is obtained by adjoining a subset of the eight edges in (2.1).

There are only 2^8 subsets. Impose degree two at each internal gadget vertex for a target cycle, or the corresponding endpoint degrees for a target path, and then contract the remaining external path segments. Connectivity and the endpoint pairing distinguish A_0, A_1, D, E. This finite case split gives exactly the cells in (6.3): each listed cell supplies a construction, and every unlisted source-target pair fails either an internal degree equation or the required contracted connectivity. Thus the table is complete.

Inspection of (6.3) shows that every source role has an output under each site. Hence every nonempty support state has a nonempty image under either relation, proving invariance of (6.4). The empty state is unavoidable and cannot belong to a nonempty robust domain, so this domain is greatest. □

The target indegree multisets of the two relations are, respectively,

$$\{1,1,1,2,2,2\}, \qquad \{1,1,1,1,2,3\}. \tag{6.5}$$

They are therefore nonconjugate even under independent permutations of source and target roles. This algebraic distinction alone does not prove that all construction histories are nonisomorphic; that issue is handled by the rooted grammar in Section 9.

7. A cube initial state

Let

$$V_0 = \{A_0, B, C, D, E\}. \tag{7.1}$$

Substitution into (6.3) gives

$$M_{\mathrm{CEN}}(V_0) = M_{\mathrm{UV}}(V_0) = V_0. \tag{7.2}$$

For the eight gadget edges, the output-interface supports obtained from V_0 are

	ua	ab	bv	xc	cd	dy	ad	bc
CEN	A_0BDE	A_0BCD	A_0CDE	A_0CDE	A_0BCE	A_0BDE	A_0C	A_0B
UV	A_0B	A_0BCE	A_0C	A_0CDE	A_0BC	A_0BDE	A_0CDE	A_0BDE.

(7.3)

Every entry is nonempty. Moreover, A_0 and B supply the traversal pair $\{D_{11}, D_{10}\}$, one of the minimal covers in (4.1).

Theorem 7.1 (stable-root criterion). Let $G_\star$ be a P_2-Hamiltonian Barnette graph with a legal ordered site. Suppose that concrete witnesses realize every role in V_0, and that every edge of $G_\star$ other than the two site edges is contained in at least one of those witnesses. Perform the first expansion at this site and thereafter choose either successor in (6.1). Then every finite history produces a P_2-Hamiltonian Barnette graph. At every marked node the available state is again V_0, and every edge retained by the next site has nonempty six-role support.

Proof. Suppose a marked parent has state V_0, with every retained edge supported by at least one concrete witness. Expand either CEN or UV. The six-traversal construction closes every source witness into a Hamilton cycle of the child while retaining every parent edge other than the two site edges. Hence every retained parent edge lies on a child Hamilton cycle. The pair A_0, B covers all eight newly installed gadget edges by Theorem 4.1; the child is therefore P_2-Hamiltonian.

Memberwise transfer through (6.3) preserves nonempty support for every edge retained by the child's marked successor site. Equation (7.3) supplies support for the retained gadget edges, and (7.2) restores the full availability state V_0. Both successor sites are legal, so Lemma 2.2 keeps the graph in the Barnette class. Induction proves the theorem. □

We now verify the hypothesis without computation. Let Q be the cube on $\{0, 1, \dots, 7\}$ with edge set

$$\{01, 03, 04, 12, 17, 23, 26, 35, 45, 47, 56, 67\}. \tag{7.4}$$

Use the facial square $(0, 1, 2, 3)$ and the opposite edges

$$(u, v, x, y) = (0, 1, 2, 3), \qquad p = 01, \qquad q = 23. \tag{7.5}$$

The site is legal: its endpoints are distinct, u, x lie in the same bipartition class, and the residual facial paths $1 - 2$ and $3 - 0$ have odd length.

Consider the following three Hamilton cycles and two spanning paths:

$$\begin{aligned} A_0 &: 0-1-2-3-5-6-7-4-0, \\ B &: 0-1-2-6-7-4-5-3-0, \\ C &: 0-3-2-1-7-6-5-4-0, \\ D &: 1-7-4-0-3-5-6-2, \\ E &: 0-4-7-1-2-6-5-3. \end{aligned} \tag{7.6}$$

The first cycle contains p, q; deleting them leaves the pairing $(03)(12) = (uy)(vx)$, so its role is A_0. The cycles B, C have roles C_{10}, C_{01}, respectively. The paths D, E avoid p, q, span Q, and have endpoints v, x and u, y.

The cycles A_0, B, C cover every edge in (7.4). Including the two spanning-path roles, the exact five-role supports are

e	01	03	04	12	17	23
$S(e)$	A_0B	BCD	A_0CDE	A_0BCE	CDE	A_0C

e	26	35	45	47	56	67
$S(e)$	BDE	A_0BDE	BC	A_0BDE	A_0CDE	A_0BC

(7.7)

Thus Q is P_2-Hamiltonian, and every retained edge has nonempty support in V_0.

Corollary 7.2 (cube family). Starting from the cube at (7.5), every finite CEN/UV history is a P_2-Hamiltonian Barnette graph. Its depth k member has $8 + 4k$ vertices.

Proof. The five explicit sequences in (7.6) verify every hypothesis of Theorem 7.1. □

8. Global ordered-edge separation

The preceding support invariant proves that every edge is Hamiltonian, but the stronger ordered contain-one/avoid-one property requires two additional pieces of information. Each retained edge needs a witness that contains it and a witness that avoids it, and each ordered pair of retained edges needs a single witness that separates the two incidences.

Write

$$\mathcal{T} = \{A, B, C, D, E, L\},$$

where $A = A_0$ is the C_{11} pairing $(uy)(vx)$, $L = A_2$ is the two-path linkage with pairing $(uv)(xy)$, and B, C, D, E are the four roles in Table 1. For a concrete family of root witnesses and a retained edge r, put

$$P(r) = \{\tau \in \mathcal{T} : \text{some role-}\tau \text{ witness contains } r\},$$
$$N(r) = \{\tau \in \mathcal{T} : \text{some role-}\tau \text{ witness avoids } r\}.$$

For an ordered pair of distinct retained edges define

$$J(r, s) = \{\tau \in \mathcal{T} : \text{some single role-}\tau \text{ witness contains } r \text{ and avoids } s\}. \tag{7.8}$$

The joint set is stronger than the two marginal sets.

Use X/Y for the pair $(P, N) = (X, Y)$. The following 22-state domain is the compact closed domain for the two standard successor sites:

$$\begin{aligned}\Sigma = \{&ABC/ADEL, ABCD/AEL, ABCE/ADL, ACDEL/AB, ABDEL/AC,\\ &ABCDEL/ABCDEL, ABCDEL/ABCEL, ABCDEL/ABDEL, ABCDEL/ABEL,\\ &ABCDEL/ADEL, ABCDEL/AEL,\\ &ABCEL/ABCDEL, ABCEL/ABDEL, ABCEL/ADEL,\\ &ABDEL/ABCDEL, ABDEL/ABCEL, ABDEL/ACDEL, ABDEL/ACEL,\\ &ACDEL/ABDEL, ACDEL/ABEL, ADEL/ABCDEL, ADEL/ABCEL\}.\end{aligned} \tag{7.9}$$

To include the reflected ordered-terminal convention, write $\sigma(u, v, x, y) = (x, y, u, v)$ and let $\rho = (B\ C)(D\ E)$, fixing A and L. The direct reflected relations (target ← source) are

	A	B	C	D	E	L
CEN_σ	BCL	D	E	A	A	A
UV_σ	ABL	D	E	A	C	A

(7.10)

The map σ is a cyclic shift of the same facial order, and it preserves the condition that the first and third terminals lie in the same bipartition class. Thus the reflected choices use the same two unoriented edge pairs and are legal whenever

the corresponding standard choice is legal. The standard relations are represented by the rows in (7.13)–(7.14) below. The reflected maps do not preserve Σ itself: for example,

$$ADEL/ABCEL \mapsto ABCDEL/ACDEL, \qquad ABC/ADEL \mapsto ADEL/ABCDL \tag{7.11}$$

under CEN_σ and UV_σ, respectively. Define

$$\Sigma^* := \mathrm{cl}_{\mathrm{CEN},\mathrm{UV},\mathrm{CEN}_\sigma,\mathrm{UV}_\sigma}(\Sigma). \tag{7.12}$$

The exact finite closure has $|\Sigma^*| = 40$, with 18 states added to Σ. This is the least forward closure containing the displayed seed domain; it is not a claim of global minimality under a different state encoding.

For reproducibility, the complete closure (using the role order A, B, C, D, E, L) is

$$\begin{gathered}
ABC/ADEL,\ ABCD/AEL,\ ABCE/ADL,\ ABDEL/AC,\\
ACDEL/AB,\ ADEL/ABCDL,\ ADEL/ABCEL,\ ABCDL/ADEL,\\
ABCEL/ADEL,\ ABDEL/ACDL,\ ABDEL/ACEL,\ ACDEL/ABDL,\\
ACDEL/ABEL,\ ABCDEL/ADL,\ ABCDEL/AEL,\ ADEL/ABCDEL,\\
ABCDL/ABDEL,\ ABCDL/ACDEL,\ ABCEL/ABDEL,\ ABCEL/ACDEL,\\
ABDEL/ABCDL,\ ABDEL/ABCEL,\ ABDEL/ACDEL,\ ACDEL/ABCDL,\\
ACDEL/ABCEL,\ ACDEL/ABDEL,\ ABCDEL/ABDL,\ ABCDEL/ABEL,\\
ABCDEL/ACDL,\ ABCDEL/ACEL,\ ABCDEL/ADEL,\ ABCDL/ABCDEL,\\
ABCEL/ABCDEL,\ ABDEL/ABCDEL,\ ACDEL/ABCDEL,\ ABCDEL/ABCDL,\\
ABCDEL/ABCEL,\ ABCDEL/ABDEL,\ ABCDEL/ACDEL,\ ABCDEL/ABCDEL.
\end{gathered} \tag{7.12a}$$

The list is regenerated from the complete local enumeration by the script in the supplementary material.

The exact local transfer rows are recorded by their incidence word on the six edges retained by the next marked site. In the CEN order (ua, ab, bv, xc, cd, dy), the nonempty rows are

source	$A \to D$	$A \to E$	$A \to L$	$B \to A$	$C \to A$	$D \to C$	$E \to B$	$L \to A$
word	111101	101111	101101	101010	010101	011110	110011	101101.

(7.13)

In the UV order (ab, xc, cd, dy, ad, bc), they are

source	$A \to A$	$A \to E$	$A \to L$	$B \to A$	$C \to D$	$D \to C$	$E \to B$	$L \to A$
word	111100	110111	010111	001011	010111	111010	101101	010111.

(7.14)

These tables are complete: they come from all 2^8 gadget subsets after imposing the relevant degree, connectivity and endpoint conditions. The reflected tables in (7.10) have the same local incidence words, with their target labels relabelled by $\rho = (B\ C)(D\ E)$.

Lemma 8.1 (40-state reflected signed closure). For each of the four standard or reflected maps in (7.13)–(7.10), applying the memberwise rows to either coordinate of every state in Σ^* gives a state in Σ^*. For every $X/Y \in \Sigma^*$ and every retained position, some transition from a role in X has local bit 0, and some transition from a role in Y has local bit 1. The final closure rows

	ua	ab	bv	xc	cd	dy	ad	bc
A	1	1	1	1	1	1	0	0
B	1	0	1	0	1	0	1	1
C	0	1	0	1	0	1	1	1
D	0	1	1	1	1	0	1	0
E	1	1	0	0	1	1	0	1
L	1	0	1	1	0	1	1	1

(7.15)

separate every ordered pair of distinct final gadget edges. For a final position j, let Z_j and O_j be the sets of roles whose row in (7.15) has bit 0, respectively 1, at j:

j	ua	ab	bv	xc	cd	dy	ad	bc
Z_j	CD	BL	CE	BE	CL	BD	AE	AD
O_j	$ABEL$	$ACDE$	$ABDL$	$ACDL$	$ABDE$	$ACEL$	$BCDL$	$BCEL$

(7.15a)

Then every $X/Y \in \Sigma^*$ satisfies $X \cap Z_j \neq \varnothing$ and $Y \cap O_j \neq \varnothing$ for all eight final positions j. The six retained positions born in a nonfinal layer are also separated by a local row with bits 10 in every ordered pair of positions (30 ordered pairs for each of the four maps).

Proof. This is a finite Boolean check on the displayed rows and the four reflected relations: take the two coordinate images of each state in Σ^*, check the six retained-position conditions, and inspect the final columns. The reflected relation and all local incidence rows are obtained from the complete 2^8 local subset enumeration, not from a bounded-depth graph search. The same enumeration checks the 30 ordered $(1,0)$ separations among the six retained positions in each map; these are the joint-witness conditions for two edges born in the same nonfinal annulus. For the final layer, the enumeration shows that a source of each role τ has exactly one completion to a Hamilton cycle, with gadget incidence word the row τ of (7.15). The 56 ordered final pairs and the $2 \cdot 8 \cdot 40 = 640$ conditions (7.15a) are then read off these six rows. □

For the cube seed, the signed states of edges born in a nonfinal layer are

$$\begin{array}{c|l} \mathrm{CEN} & ua, dy : ABDEL/AC; \quad ab : ABCD/AEL; \quad bv, xc : ACDEL/AB; \quad cd : ABCE/ADL \\ \mathrm{UV} & ab : ABCE/ADL; \quad xc, ad : ACDEL/AB; \quad cd : ABC/ADEL; \quad dy, bc : ABDEL/AC \end{array} \tag{7.16}$$

and every entry belongs to Σ^*. For the reflected sites the corresponding birth states are

$$\begin{array}{c|l} \mathrm{CEN}_\sigma & ua : ACDEL/AB; \quad ab : ABCE/ADL; \\ & bv, xc : ABDEL/AC; \quad cd : ABCD/AEL; \quad dy : ACDEL/AB \\ \mathrm{UV}_\sigma & ab : ABCD/AEL; \quad xc : ABDEL/AC; \quad cd : ABC/ADEL; \\ & dy : ACDEL/AB; \quad ad : ABDEL/AC; \quad bc : ACDEL/AB \end{array} \tag{7.17}$$

and all of these entries belong to Σ^*.

Theorem 8.2 (portable seed-signature transfer). Let a legal ordered site in a Barnette graph have six nonempty concrete role families A, B, C, D, E, L. Assume that every retained root edge r has $(P(r), N(r)) \in \Sigma^*$, and that $J(r,s) \neq \varnothing$ for every ordered pair of distinct retained root edges. Assume also that the chosen standard or reflected CEN/UV successors remain legal Barnette sites. Then every positive-depth descendant obtained by an arbitrary finite word in the four-map alphabet

$$\{\mathrm{CEN}, \mathrm{UV}, \mathrm{CEN}_\sigma, \mathrm{UV}_\sigma\}$$

has global H^{+-}: for every ordered pair of distinct edges (e, f), a Hamilton cycle contains e and avoids f. If the empty word is included, the root itself must additionally have global H^{+-}.

Proof. Proceed by induction over the annuli. For two older retained edges, the same witness supplied by $J(r,s)$ is carried through a nonempty source-to-target row, and all older incidences are unchanged. If both edges were born in the same nonfinal layer, the 10-row assertion of Lemma 8.1 gives the separation at birth, after which the same continuation applies.

For an older edge r and a gadget edge j of a nonfinal layer, use $P(r)$ and the bit-0 assertion; in the reverse order, use $N(r)$ and the bit-1 assertion. If j lies in the final gadget, a role-τ witness containing r with $\tau \in P(r) \cap Z_j$ closes to a Hamilton cycle that contains r and avoids j, and a role-τ' witness avoiding r with $\tau' \in N(r) \cap O_j$ closes to one that contains j and avoids r; both intersections are nonempty by (7.15a). The signed state remains in Σ^* after each nonfinal layer, and the explicit birth rows initialize new retained edges. If both edges lie in the final gadget, (7.15) supplies a source role and a row with bits 10; role availability makes that row concrete. The four blocks cover every ordered pair of distinct edges. □

For the cube, three A-role cycles (the cycle $A^{(2)}$ is the cycle A_0 of (7.6)) and one L-linkage have supports (after deleting $01, 23$)

$$\begin{array}{c|l} A^{(1)} & \{03, 17, 26, 45, 47, 56\} \\ A^{(2)} & \{04, 12, 35, 47, 56, 67\} \\ A^{(3)} & \{04, 17, 26, 35, 45, 67\} \\ L & \{04, 17, 26, 35, 47, 56\}. \end{array} \tag{7.18}$$

Together with B, C, D, E in (7.6), these are eight distinct concrete supports. Their retained-edge incidence columns form an antichain, so for every ordered pair of distinct retained edges one support contains the first and avoids the second. The cube therefore satisfies all seed conditions.

Corollary 8.3 (cube global H^{+-}).

Every finite cube word in the standard or reflected CEN/UV successors has global H^{+-}. In particular, every standard CEN/UV word, and every such graph, is P_2-Hamiltonian.

Proof. For the empty word, the root cube itself is covered by the six Hamilton cycles

$$\begin{array}{lll} 01235674, & 01267453, & 01745623, \\ 01762354, & 03217654, & 03562174, \end{array}$$

where each displayed sequence is closed by its final edge back to 0. Their edge-incidence columns have the $1,0$ pattern for every ordered pair of distinct cube edges, so the root has global H^{+-}. For every positive depth, the explicit supports above verify the seed conditions and Theorem 8.2 supplies the induction. □

9. The ordered-boundary grammar

Theorem 7.1 gives a binary history tree, but it does not count unlabelled graphs. We now classify the corresponding local patches with the old ordered boundary fixed pointwise.

Let $\mathcal{B}$ be the basic four-pole consisting of a square

$$a - b - c - d - a$$

with ordered boundary attachments

$$U - a, \qquad V - b, \qquad X - c, \qquad Y - d. \tag{8.1}$$

For an ordered four-pole N, write $n_T(N)$ for the internal neighbour of terminal $T \in \{U, V, X, Y\}$.

Define two outer-annulus operators F_0, F_1. For $F_0(N)$, introduce outer vertices a, b, c, d, use boundary paths

$$U - a - b - V, \qquad X - c - d - Y,$$

and attach N by

$$bn_U(N), \qquad cn_V(N), \qquad dn_X(N), \qquad an_Y(N). \tag{8.2}$$

The only internal edges of this outer block are ab, cd.

For $F_1(N)$, the outer vertices induce $a - b - c - d - a$, with

$$X - c, \qquad Y - d, \qquad an_V(N), \qquad bn_X(N), \tag{8.3}$$

while the U- and Y-terminals of N become the new outer U- and V-terminals. These are precisely the annuli left by expanding the CEN and UV sites.

Encode CEN by 0 and UV by 1. If $b_1 \cdots b_k$ is a depth-k history, then the first expansion installs $\mathcal{B}$, the letter b_i chooses the site expanded at the next step, and the unmarked depth-k patch is

$$F_{b_1} \circ F_{b_2} \circ \cdots \circ F_{b_{k-1}}(\mathcal{B}). \tag{8.4}$$

The final letter b_k only marks the site for a future expansion. For a composition word $w = i_1 \cdots i_m$ (with each $i_j \in \{0, 1\}$), we write

$$F_w(N) := F_{i_1} \circ \cdots \circ F_{i_m}(N), \qquad F_\varnothing(N) := N. \tag{8.4a}$$

Thus a depth-k history $b_1 \cdots b_k$ has unmarked patch $F_{b_1 \cdots b_{k-1}}(\mathcal{B})$.

The following elementary invariant makes the subsequent adjacency test intrinsic on the whole grammar, rather than only on the first few examples.

Lemma 9.1 (grammar noncollision invariant). Every pole generated from $\mathcal{B}$ by F_0, F_1 is simple and has four pairwise distinct boundary neighbours. Its X- and Y-boundary neighbours are adjacent. In every nonbasic pole, the vertices c^*, d^* defined below are distinct and uniquely determined.

Proof. The assertions are immediate for $\mathcal{B}$. In $F_0(N)$, the new boundary neighbours are the four distinct outer vertices a, b, c, d, while the X, Y neighbours are c, d and $c^* = n_V(N), d^* = n_X(N)$. In $F_1(N)$, the new boundary neighbours are $n_U(N), n_Y(N), c, d$, with the first two distinct by induction and the last two new; here $c^* = b, d^* = a$. Inner and outer vertices are disjoint, so no new edge collision occurs. In both cases cd is an edge, and the listed pairs are distinct by the induction hypothesis. □

Reading one letter

Define

$$r(N) = \mathbf{1}\big[n_V(N)n_X(N) \in E(N)\big]. \tag{8.5}$$

For the basic pole, $r(\mathcal{B}) = 1$. In $F_0(N)$, the boundary neighbours of V, X are the nonadjacent outer vertices b, c; in $F_1(N)$, they are $n_Y(N)$ and c, which are also nonadjacent. Thus

$$r(F_0(N)) = r(F_1(N)) = 0, \qquad r(N) = 1 \Longleftrightarrow N = \mathcal{B} \tag{8.6}$$

for every grammar pole.

For a nonbasic grammar pole P, let

$$c = n_X(P), \qquad d = n_Y(P).$$

The vertices c, d are adjacent and cubic. Let c^*, d^* be their unique remaining neighbours after omitting their boundary edges and the edge cd, and put

$$\kappa(P) = \mathbf{1}[c^*d^* \in E(P)]. \tag{8.7}$$

From (8.3), $c^* = b, d^* = a$, and ab is present. From (8.2), $c^* = n_V(N), d^* = n_X(N)$. Therefore

$$\kappa(F_1(N)) = 1, \qquad \kappa(F_0(N)) = r(N). \tag{8.8}$$

If $N \neq \mathcal{B}$, equations (8.6)–(8.8) give

$$\kappa(F_i(N)) = i. \tag{8.9}$$

Once i is known, the outer block is intrinsic. For $i = 0$, it is the set of the four boundary neighbours. For $i = 1$, it is $\{c, d, c^*, d^*\}$. Deleting this block recovers N with its full ordered boundary: in the F_0 case the recovered attachments are ordered by $b, c, d, a \leftrightarrow U, V, X, Y$; in the F_1 case the outer U, V recover inner terminals U, Y, and the attachments at $d^* = a, c^* = b$ recover V, X. Hence every boundary-pointwise isomorphism preserves the outer letter and induces an isomorphism of the peeled poles until the inner pole is basic.

The final cap

The final letter cannot be read, because the two caps are explicitly isomorphic. Name the F_0 outer block a, b, c, d and its inner basic square e, f, g, h. The boundary-fixed map

$$\begin{aligned} &a \mapsto e, \quad b \mapsto h, \quad c \mapsto c, \quad d \mapsto d, \\ &e \mapsto g, \quad f \mapsto b, \quad g \mapsto a, \quad h \mapsto f \end{aligned} \tag{8.10}$$

sends the internal edge set

$$\{ab, cd, be, cf, dg, ah, ef, fg, gh, he\}$$

of $F_0(\mathcal{B})$ onto that of $F_1(\mathcal{B})$. Therefore

$$F_0(\mathcal{B}) \cong F_1(\mathcal{B}) \tag{8.11}$$

by an isomorphism fixing U, V, X, Y pointwise.

Theorem 9.2 (rooted two-step delay). For composition words $s, t \in \{0, 1\}^m$ with $m \geq 1$,

$$F_s(\mathcal{B}) \cong F_t(\mathcal{B}) \quad \text{by an ordered-boundary pointwise isomorphism}$$

if and only if s and t agree in their first $m - 1$ letters (equivalently, after deleting their final letters). Hence the number of pointwise-rooted unmarked types at depth k is

$$T_k = 2^{\max(k-2,0)}. \tag{8.12}$$

Proof. If the inner pole is not basic, (8.9) recovers the outer letter from the rooted graph. The intrinsic block described above can then be peeled, preserving the full ordered boundary. Recursion forces equality of all letters except the final one. Conversely, (8.11) identifies either final letter, and applying the common outer operators extends this isomorphism through every shared prefix.

At depth $k \geq 1$, equation (8.4) uses the first $k-1$ history letters, and the last of those is invisible by (8.11); the history letter b_k only marks the next site. Thus the graph remembers exactly the first $k-2$ letters for $k \geq 2$. There is one unmarked rooted type at depths 0 and 1, yielding (8.12). □

10. From rooted types to unlabelled graphs

Let Q be the labelled cube in (7.4), and let

$$R = Q - \{01, 23\} \tag{9.1}$$

on the same eight labelled vertices. In every positive-depth descendant, the subgraph induced by the old vertices is exactly R: the first expansion deletes $01, 23$, and later operations neither delete another old–old edge nor add an old–old edge. Thus the rooted object is a graph together with a labelled injection of R whose image induces R, not an embedded copy of the complete cube Q. Attaching the boundary-pointwise patch types to this same labelled exterior therefore gives the same number of pointwise old-core rooted types: an isomorphism fixing the old labels preserves the complement of their image, and hence restricts to an isomorphism of the attached patch.

Lemma 10.1 (cubic marking bound). An unlabelled cubic n-vertex graph supports at most

$$n\,3^7 \tag{9.2}$$

pointwise-rooted types arising from this construction.

Proof. Fix a representative H of the unlabelled graph. A pointwise-rooted realization determines an injection

$$\iota : V(R) \hookrightarrow V(H)$$

that identifies every old vertex and, in particular, the ordered boundary (u, v, x, y). Since the old vertices induce R in every descendant, every valid ι is an injective graph embedding of R into H. Fix the spanning tree

$$T = \{03, 04, 45, 56, 26, 12, 17\} \subseteq E(R),$$

rooted at 0. An embedding is determined by the image of 0 and, successively along T, the choice of the image of each child among the at most three neighbours of its already embedded parent. Thus there are at most $n\,3^7$ such maps; injectivity and the remaining edges of R can only reduce this number. Rooted types supported by H are $\mathrm{Aut}(H)$-orbits of valid embeddings, so their number is no larger than (9.2). No uniqueness or recognizability of the copy of R is used. □

Proof of Theorem 1.1. By Corollary 8.3 and Theorem 9.2, depth k supplies $2^{\max(k-2,0)}$ pointwise-rooted H^{+-} Barnette types of order $8+4k$. Partition these types by their underlying unlabelled graph. Lemma 10.1 shows that every part has size at most $(8+4k)\,3^7$. Therefore

$$B_{H^{+-}}(8+4k) \geq \left\lceil \frac{2^{\max(k-2,0)}}{(8+4k)\,3^7} \right\rceil.$$

For $k = 0$, the same inequality follows directly from the certified H^{+-} cube; for $k = 1$, the positive-depth marking lemma applies and the numerator is $T_1 = 1$.

If $k \geq 2$ and $n = 8+4k$, then

$$k - 2 = \frac{n}{4} - 4,$$

so

$$B_{H^{+-}}(n) \geq \left\lceil 2^{n/4 - \log_2 n - 4 - 7\log_2 3} \right\rceil \geq 2^{n/4 - \log_2 n - 4 - 7\log_2 3} = 2^{n/4 - O(\log n)}$$

along the stated sequence of orders. □

Limitations and scope

The six-traversal criterion concerns one specified expansion. The recursive theorems concern only the two marked successor sites CEN and UV (and their reflections), not every legal site. The support conditions are sufficient for graph-level conclusions. They are necessary only inside the stated transfer systems, because other Hamilton cycles may also cover edges. The rooted classification fixes an ordered boundary, and the cube application fixes the eight old labels pointwise. The passage to unlabelled graphs uses only the coarse cubic embedding bound of Lemma 10.1, not uniqueness of the core. Consequently the constant 3^7 and the factor n are far from sharp. In the brute-force data below, the actual number of isomorphism classes at depth $k \leq 6$ is $1, 1, 1, 2, 3, 6, 10$ for $k = 0, \ldots, 6$, compared with $2^{\max(k-2,0)}$ rooted types. The comparison with Kao–Chen [15] remains open, as explained above.

Reproducibility

All proofs are written out in the text; the finite local enumerations (at most 2^8 subsets of gadget edges) can be checked by hand from the displayed tables. The supplementary archive contains independent scripts (Python standard library and `networkx`) that

- re-derive the six traversal patterns, the seven minimal covers and the triangle-free intersection graph, the cube witnesses and their supports, and the boundary-fixed cap isomorphism;
- recompute the 22-state domain, its 40-state reflected forward closure, all retained-position separation conditions and the birth states, from the complete 2^8 local enumeration;
- recompute the final Hamilton-cycle rows (7.15) from the same enumeration and check the conditions (7.15a) for all 40 states, which cover every pair of an older edge and a final gadget edge, together with all 56 ordered pairs of final gadget edges;
- check the cube seed conditions of Theorem 8.2 from the explicit witnesses;
- build every standard construction word up to depth 6 ($n \leq 32$) and every reflected word up to depth 5 ($n \leq 28$, 256 graphs). For each graph they check the Barnette conditions and verify global H^{+-} by exhaustive Hamilton-cycle enumeration. They also count isomorphism classes and old-core-fixed rooted classes; the rooted counts equal $2^{\max(k-2,0)}$ for every tested depth.

These computations test finitely many cases and are not a proof; the theorems rest on the written arguments.

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